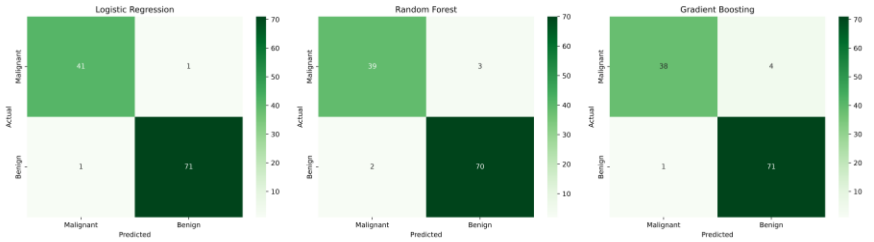
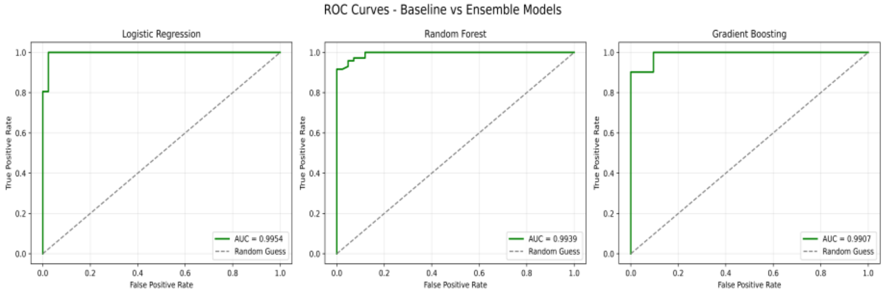
Ensemble Learning, Model Optimization & Real-World ML Pipeline

AI & Machine Learning — Task 5

Model Comparison — Baseline vs Ensembles

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9939
Gradient Boosting	0.9561	0.9467	0.9861	0.9660	0.9907





- All three models discriminate classes very well (AUC > 0.99 for most)
- The linear baseline was already near-perfect on this dataset

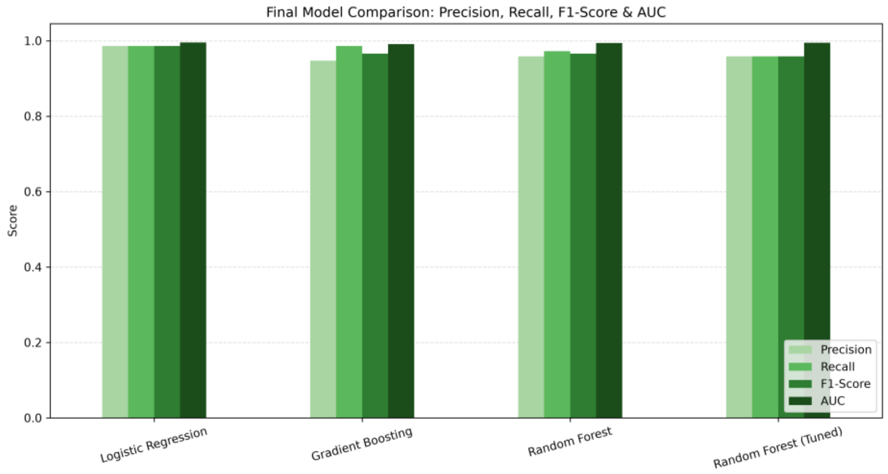
Hyperparameter Tuning (GridSearchCV)

- Ensembles showed overfitting risk: Train Acc 100% vs Test Acc 95.6% (gap 0.044) for both Random Forest & Gradient Boosting
- Tuned Random Forest via GridSearchCV, 5-fold CV, scored on F1
- Grid: n_estimators, max_depth, min_samples_split
- **Best Parameters: max_depth=7, min_samples_split=5, n_estimators=50**
- **Best CV F1-Score: 0.9685**

Effect of Tuning

Model	Accuracy	F1-Score	AUC
Random Forest (untuned)	0.9561	0.9655	0.9939
Random Forest (tuned)	0.9474	0.9583	0.9947

- Tuning improved AUC slightly but did not improve test F1/Accuracy on this split
- Confirms cross-validated tuning must be checked against held-out results, not assumed to always help



Final Model Decision

Selected Model: Logistic Regression (baseline)

- Highest Accuracy (0.9825), F1-Score (0.9861), and AUC (0.9954) of every model tested — tuned or untuned
- Most stable (train-test gap 0.007 vs 0.044 for ensembles) and most interpretable
- Ensembles + GridSearchCV tuning were implemented and validated but did not beat the simpler linear model here
- Conclusion: always benchmark ensembles/tuning against the baseline — added complexity should earn its place with real results